

EXISTENCE OF PERIODIC SOLUTIONS TO EQUATIONS WITH MEASURES VIA THE SHOOTING METHOD

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ABSTRACT. In this paper, we investigate the existence of periodic solutions for systems of measure differential equations (MDEs). Our approach is based on the shooting method, which allows us to establish a theoretical result guaranteeing the existence of solutions to the problem under consideration. Moreover, the shooting method serves as a robust tool for the numerical approximation of such systems. In the final section, we apply this approach to demonstrate the existence of periodic solutions for a forced pendulum subject to impulsive forces. Additionally, we present a computational implementation to obtain approximate solutions.

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1. INTRODUCTION AND MAIN THEOREM

The goal of this paper is to establish the existence of periodic solutions to *Measure Differential Equations* (briefly, MDEs). More specifically, we seek solutions to the following boundary value problem

$$(1.1) \quad \begin{cases} d\varphi = F(t, \varphi(t)) \, d\nu \\ \varphi(0) - \varphi(T) = 0, \end{cases}$$

where $\nu : \mathcal{B}([0, T]) \rightarrow \mathbb{R}^m$ is a vector measure (see [10]) with finite total variation, $\mathcal{B}([0, T])$ is the σ -algebra of all Borel measurable subsets of $[0, T]$, and $F : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ is a matrix-valued Carathéodory function with respect to ν . That is, $F(t, x)$ is Borel measurable as a function of t for every $x \in \mathbb{R}^{n \times m}$, and continuous as a function of x for $|\nu|$ -a.e. $t \in [0, T]$.

If $I \subset \mathbb{R}$ is an interval, in what follows we denote by $BV(I, \mathbb{R}^n)$ the set of all $\mathbb{R}^n$ -valued functions of bounded variation, and by μ_φ the Lebesgue–Stieltjes measure induced by φ (see [8, 12, 13, 17, 26]). Whenever $f \in L^1(I, \mu_\varphi)$, for simplicity we write $\int f \, d\varphi$ instead of $\int f \, d\mu_\varphi$.

In the literature, several notions of solutions for MDEs have been proposed, which in some cases lead to non-equivalent results for the same equation (see [31]). In this paper, we adopt the definition introduced in [5].

Definition 1.1 (see [5]). *We say that $\varphi \in BV([0, t_\varphi], \mathbb{R}^n)$ is a solution of $d\varphi = F(t, \varphi(t)) \, d\nu$ if φ is left-continuous and, for every Borel measurable*

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set $A \subset [0, t_\varphi)$, the following holds:

$$(1.2) \quad \int_A d\varphi = \int_A F(s, \varphi(s)) d\nu.$$

We say that φ is a solution of problem (1.1) if φ solves $d\varphi = F(t, \varphi(t)) d\nu$ on $[0, T)$, is left-continuous on $[0, T]$, and satisfies $\varphi(0) - \varphi(T) = 0$.

Many of the MDEs considered in the literature are particular instances of (1.1). For example, it is common to encounter systems of MDEs written in the form

$$d\varphi = f(t, \varphi) + g(t, \varphi) d\mu,$$

where μ is a real-valued signed Borel measure and f and g are vector-valued functions (see [7]). Clearly, this equation can be viewed as a particular case of (1.1) by taking $F = (f, g)$ and $\nu = (\lambda, \mu)^t$, where λ denotes the Lebesgue measure on $\mathbb{R}$.

The Radon–Nikodym theorem allows us to rewrite equations involving vector measures in terms of positive measures. Let ν be a vector measure with values in $\mathbb{R}^m$. Then there exists a function $H \in L^1(|\nu|; \mathbb{R}^m)$ such that, for every measurable set E , the following representation holds:

$$\nu(E) = \int_E H(s) d|\nu|.$$

As a consequence, any solution of (1.1) is also a solution of

$$(P) \quad \begin{cases} d\varphi = f(t, \varphi(t)) d\mu \\ \varphi(0) - \varphi(T) = 0, \end{cases}$$

where $f : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the vector field defined by $f = FH$, and μ is a positive measure. Accordingly, throughout this paper we focus our analysis on problem (P). Depending on the measure ν , the function f may be discontinuous with respect to time, even if F is continuous with respect to that variable. This observation highlights the importance of not imposing, or imposing only in the weakest possible sense, continuity assumptions on f with respect to t .

MDEs appear in nonsmooth models across several branches of science, ranging from the motion of molecules in ideal gases to the dynamics of impacts between a ball and a racket. In physics and mechanical engineering, MDEs arise in the study of granular matter and sandpiles; in robotics, they are used to model impacts at joints (see [7] and [21]). A particular case in which MDEs arise is given by impulsive models; see [2, 4, 20].

For example, in mechanical systems, the impact between two bodies can be regarded as a short-duration phenomenon that produces a sudden change in the system dynamics. Consequently, impacts are usually represented as large forces acting over an infinitesimally short time interval. Control theory provides another source of equations involving measures. For instance, in the optimization of space flight mechanics and in robotic motion, classical variational methods do not allow one to obtain optimal controls with an impulsive character (see [31]).

Beyond applications, differential equations with measures constitute, from a purely mathematical point of view, a natural generalization of ordinary differential equations, and, in our opinion, this fact justifies their study.

The problem of finding periodic solutions to impulsive systems has attracted considerable attention. It has been addressed in the literature using various techniques; see, for example, [9, 15, 16, 18, 22–25, 29, 30, 32]. To the best of our knowledge, the present paper is the first to approach this problem via the shooting method.

Several integration theories have been used to deal with problems such as those described above, namely the Lebesgue–Stieltjes, Perron–Stieltjes, and Kurzweil–Henstock integrals (see, for example, [6, 19, 27, 28]).

We employ the shooting method to find solutions to the periodic boundary value problem associated with a first-order system of differential equations. The idea consists of seeking a solution u_α to the initial value problem

$$(IVP) \quad \begin{cases} u' = f(t, u(t))d\mu \\ u(0) = \alpha, \end{cases}$$

and then finding a value $\alpha \in \mathbb{R}^2$ such that $u_\alpha(T) = \alpha$. That is, α is a fixed point of the Poincaré operator $P : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, defined by $P(\alpha) = u_\alpha(T)$. Under suitable conditions, an application of Brouwer’s Theorem ensures the existence of a solution to the periodic problem.

Our main result is the following

Theorem 1.2. *Let μ be a finite positive Borel measure on $[0, T]$. Let $\Omega \supset [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an open set. Let $f : \Omega \rightarrow \mathbb{R}^n$ be a function satisfying the following conditions:*

- (H₁) *For each $x \in \mathbb{R}^n$, $f(t, x)$ is μ -measurable with respect to the variable t .*
- (H₂) *The function f is Lipschitz with respect to the variable x .*
- (H₃) *f is continuous with respect to t at every point such that $\mu(\{t\}) = 0$.*
- (H₄) *There exists a ball $B(0, R)$ such that:*
 - (H₄₁) *$f(\cdot, x^*) \in L^2([0, T], d\mu)$ for some $x^* \in B(0, R)$.*
 - (H₄₂) *For every $t \in [0, T]$ and every $x \in \overline{B(0, R)}$, we have*

$$x + f(t, x)\mu(\{t\}) \in \overline{B(0, R)}.$$

- (H₄₃) *For every $t \in [0, T]$ with $\mu(\{t\}) = 0$ and every x such that $|x| = R$, we have $f(t, x) \cdot x < 0$.*

Then, problem (P) has at least one solution.

The paper is organized as follows. In Section 2 we review some definitions and results on Lebesgue–Stieltjes measures for functions of bounded variation. In Section 3 we prove a version of Gronwall’s lemma involving inequalities with respect to a positive Borel measure instead of the Lebesgue measure. This lemma will be used to prove the main theorem in Section 4. At the end of Section 4, we present a generalization of Theorem 1.2 (see Theorem 4.8). Finally, in Section 5, we apply our result to the problem of finding periodic solutions to a system describing a pendulum subject to an impulsive force. We also present a computational implementation of the shooting method to approximate the corresponding solution.

2. PRELIMINARIES

As usual, we denote by $\mathbb{R}^n$ the set of all n -tuples with real components. If $x \in \mathbb{R}^n$, we write the Euclidean norm as $|x| = \sqrt{x_1^2 + \cdots + x_n^2}$, and the taxicab

norm as $|x|_1 = |x_1| + \cdots + |x_n|$. We use the dot product on $\mathbb{R}^n$ defined by $x \cdot y = x_1 y_1 + \cdots + x_n y_n$.

The set of all continuous functions $f : [0, T] \rightarrow \mathbb{R}^n$ will be denoted by $C([0, T], \mathbb{R}^n)$, or simply $C([0, T])$ when the codomain is $\mathbb{R}$.

We briefly introduce some notation related to measure theory; we refer to [8, 12, 13, 17, 26] for further details on the material presented in this section.

Given a Borel measurable subset A of $\mathbb{R}$, we denote by $\mathcal{B}(A)$ the σ -algebra of all Borel subsets of A . If μ is a signed measure, we write $\mu = \mu^+ - \mu^-$ for the Jordan decomposition of μ , and $|\mu|$ for the *total variation* of μ , that is, $|\mu| = \mu^+ + \mu^-$. We write $\mu \perp \eta$ to indicate that μ and η are *mutually singular* measures. We say that a measure μ is *continuous* if, for every $t \in \Omega$, it holds that $\mu(\{t\}) = 0$. We use the symbol λ to denote the Lebesgue measure on $\mathbb{R}$.

As mentioned above, the set of all functions of bounded variation $\varphi : [0, T] \rightarrow \mathbb{R}^n$ is denoted by $BV([0, T], \mathbb{R}^n)$. If $\varphi \in BV([0, T], \mathbb{R}^n)$, we denote by $V(\varphi, [0, T])$ the *total variation* of φ . As is well known, functions in $BV([0, T], \mathbb{R}^n)$ have one-sided limits at every $t \in [0, T]$ and are continuous except at a set that is at most countable. If $\varphi \in BV([0, T], \mathbb{R}^n)$, then there exists a unique Borel measure, denoted by μ_φ , such that $\mu_\varphi([a, b)) = \varphi(b^-) - \varphi(a^-)$ for every a and b in $[0, T]$. This measure is called the *Lebesgue–Stieltjes measure induced by φ* (see [13] for details). We will usually assume that φ is left-continuous, in which case $\mu_\varphi([a, b)) = \varphi(b) - \varphi(a)$.

The existence of local and global solutions to initial value problems for MDEs has been studied extensively in the literature. We refer, in particular, to the approach developed in [5], where the following theorem is established. We adapt its statement to suit the needs of this article.

Theorem 2.1. *Let μ be a finite measure, Ω be an open set of $\mathbb{R} \times \mathbb{R}^n$, and $(0, \alpha) \in \Omega$. Suppose that $f : \Omega \rightarrow \mathbb{R}^n$ is locally Lipschitz with respect to the second variable for $t \geq 0$, and that φ is a solution of (IVP) defined on a maximal half-open interval $[0, t_1)$. Then exactly one of the following assertions holds:*

- (1) *For every $K \subseteq \Omega$ there exists $t_2 \in [0, t_1)$ such that $(t, \varphi(t)) \notin K$ for all $t \in (t_2, t_1)$,*
- (2) *The limit $x_1 := \lim_{t \rightarrow t_1^-} \varphi(t)$ exists, $(t_1, x_1) \in \Omega$, and $(t_1, x_1 + f(t_1, x_1)\mu(\{t_1\})) \notin \Omega$.*

If $|\mu|(\{t_1\}) = 0$, then alternative 1 holds. Otherwise, either of the two assertions may occur.

3. A GRONWALL-TYPE INEQUALITY FOR MEASURES

In this section, we introduce a Gronwall-type inequality. While preparing this manuscript, we became aware that inequalities of this type involving measures had already been studied in the literature (see, for example, [11, Th. 5.1] and [3, Th. 17.4]). Nevertheless, we include our proof, as it differs from previously known approaches and also provides certain improvements over existing results; see Remark 3.4 below.

We first prove a Gronwall-type inequality for continuous measures and then extend this result to arbitrary positive Borel measures.

Lemma 3.1. *Let μ be a finite, positive, continuous measure. If $u \in L^1(\mu)$ satisfies*

$$(3.1) \quad u(t) \leq c + \int_{[0,t)} u(r) d\mu,$$

then $u(t) \leq ce^{\mu([0,t))}$.

Proof. We define $w(t) = c + \int_{[0,t)} u(r) d\mu(r)$, $F(s) = -e^{-s}$, and $h(t) = \mu([0,t))$. Note that h is a continuous function and $\mu_h = \mu$. The Lebesgue–Stieltjes measure μ_w associated with w satisfies $\mu_w \ll \mu$ and $d\mu_w = u d\mu$. Additionally, using the chain rule for BV functions (see [1, Th. 3.99]; note that μ has no jump part), we have that $\mu_{F \circ h} = F'(h(r)) \mu$. Therefore, we obtain

$$\int_{[0,t)} -F(h(r)) dw \leq \int_{[0,t)} F'(h(r)) w(r) d\mu = \int_{[0,t)} w(r) d\mu_{F \circ h}.$$

Then, using the integration by parts formula (see [26, Th. 14.1]) and taking into account the continuity of $F \circ h$, we have

$$\begin{aligned} 0 &\leq \int_{[0,t)} F(h(r)) dw + \int_{[0,t)} w(r) \mu_{F \circ h} = \mu_{(F \circ h)w}([0, t)) \\ &= F(h(t))w(t) - F(h(0))w(0). \end{aligned}$$

Therefore, we conclude

$$u(t) \leq w(t) \leq c \exp(\mu([0, t))).$$

□

Definition 3.2. *Let μ be a finite positive measure and let $D = \{\tau \in [0, T] \mid \mu(\{\tau\}) > 0\}$.*

For any Borel set A , we define the jump part of μ as the measure μ_j induced by μ on D , that is,

$$\mu_j(A) := \sum_{\tau \in D \cap A} \mu(\{\tau\}) = \mu(D \cap A).$$

Moreover, we introduce the continuous measure

$$\bar{\mu}(A) := \mu(A) - \mu_j(A) = \mu(A \setminus D).$$

In other words, in the terminology of Ambrosio et al. (see [1]), $\bar{\mu}$ is the sum of the Cantor and absolutely continuous parts of μ . Note that μ_j and $\bar{\mu}$ are mutually singular Borel measures.

Theorem 3.3 (Generalized Gronwall Inequality). *Let μ be a finite, positive measure. If $u \in L^1(\mu)$, $u \geq 0$, and u satisfies*

$$(3.2) \quad u(t) \leq c + \int_{[0,t)} u(r) d\mu,$$

then

$$(3.3) \quad u(t) \leq cK(t)e^{K(t)\bar{\mu}([0,t))},$$

where $K(t) = \prod_{\tau \in D \cap (0,t)} (1 + \mu(\{\tau\}))$.

Proof. We fix $\epsilon > 0$. By the absolute continuity of the integral, there exists $\delta > 0$ such that

$$(3.4) \quad \text{if } \mu(A) \leq \delta, \quad \text{then} \quad \left| \int_A u(r) d\mu \right| \leq \epsilon.$$

We define $w(t) = c + \int_{[0,t)} u(s) d\mu$. Fix an arbitrary $t \in [0, T)$. Since D is countable, we may write $D \cap [0, t) = \{t_n\}_{n=1}^\infty$. Moreover, since

$$(3.5) \quad \sum_{n=1}^\infty \mu(\{t_n\}) = \mu(D \cap [0, t)) < \infty,$$

there exists $N \in \mathbb{N}$ such that

$$(3.6) \quad \sum_{n=N+1}^\infty \mu(\{t_n\}) \leq \delta.$$

Let $\{s_1, \dots, s_N\}$ be the increasing reordering of the set $D_\delta := \{t_1, \dots, t_N\}$. Additionally, we set $s_{N+1} = t$.

Decomposing the integral and using the inequality $u \leq w$, we obtain

$$(3.7) \quad \begin{aligned} w(s_{j+1}) &\leq w(s_j) [1 + \mu(\{s_j\})] + \int_{(s_j, s_{j+1})} u(s) d\mu, \\ &\leq w(s_j) [1 + \mu(\{s_j\})] + K(t) \int_{(s_j, s_{j+1})} u(s) d\mu \end{aligned}$$

for $j = 1, \dots, N$.

By an inductive argument, and using that $\mu = \bar{\mu}$ on $[0, t) \setminus D$, together with (3.4), (3.6), and the fact that $\bar{\mu}(D) = 0$, we obtain

$$\begin{aligned} w(t) &\leq cK(t) + K(t) \int_{[0,t) \setminus \{t_1, \dots, t_N\}} u(s) d\mu \\ &= cK(t) + K(t) \int_{[0,t) \setminus D} u(s) d\mu + K(t) \int_{D \setminus \{t_1, \dots, t_N\}} u(s) d\mu \\ &\leq cK(t) + K(t)\epsilon + K(t) \int_{[0,t)} u(s) d\bar{\mu} + \epsilon. \end{aligned}$$

Letting $\epsilon \rightarrow 0$, and taking into account that $K(r)$ is increasing for every $r \in [0, t]$, we obtain

$$u(r) \leq w(r) \leq cK(t) + K(t) \int_{[0,r)} u(s) d\bar{\mu}.$$

Now, since $\bar{\mu}$ is a continuous measure, we are in a position to apply Lemma 3.1. Therefore, we obtain (3.3). $\square$

Remark 3.4. In [11, Th. 5.1], a Gronwall inequality is proved by an elegant argument involving the iteration of (3.2). In some situations, our inequality may yield a sharper estimate than [11, Th. 5.1]. For instance, if μ is a purely

atomic measure with discontinuity points $\{t_n\}_{n \in \mathbb{N}}$ in $[0, t]$, then (3.2) yields the estimate

$$u(t) \leq c \prod_{j=1}^{\infty} (1 + \mu(\{t_j\})).$$

On the other hand, [11, Th. 5.1] gives the somewhat weaker bound

$$u(t) \leq ce^{\sum_{j=1}^{\infty} \mu(\{t_j\})}.$$

In [3, Th. 17.4], a Gronwall inequality is obtained under the assumption that the discontinuity points of μ are isolated. We observe that our argument allows accumulation points in the discontinuity set.

Since

$$K(t) \leq \exp \left(\sum_{s \in D \cap [0, t]} \mu(\{s\}) \right) \leq \exp(K(t)\mu_a([0, t])),$$

inequality (3.3) implies that

$$(3.8) \quad u(t) \leq ce^{K(t)\mu([0, t])}.$$

This estimate is sufficient for our purposes.

4. PROOF OF MAIN THEOREM

Throughout this section, we assume that f is as in the statement of Theorem 1.2.

Proposition 4.1. *Let φ_α be a solution of (IVP). Then $[0, T] \subset [0, t_\varphi]$ and $\varphi_\alpha \in L^\infty([0, T], \mathbb{R}^n)$. Moreover, there exists a function $0 \leq b \in L^2([0, T], d\mu)$ such that, if $s \leq t$, then*

$$|\varphi_\alpha(t) - \varphi_\alpha(s)| \leq \int_{[s, t]} b(r) d\mu(r).$$

If μ is continuous at $t_0 \in [0, T]$, then φ_α is continuous at t_0 .

Proof. As a consequence of (H_2) and (H_{41}) , we obtain

$$(4.1) \quad |f(t, x)| \leq |f(t, x^*)| + L|x - x^*| = \gamma(t) + L|x|,$$

where L is the Lipschitz constant of f and $\gamma(t) = f(t, x^*) + L|x^*| \in L^2([0, t], \mu)$. Therefore, using (1.2) with $A = [0, t]$, we get

$$|\varphi_\alpha(t)| \leq C + L \int_{[0, t]} |\varphi_\alpha(s)| d\mu.$$

where $C = |\varphi_\alpha(0)| + \|\gamma\|_{L^1(\mu)}$. Applying inequality (3.8), we deduce that, for every $t \in [0, t_\varphi]$,

$$|\varphi_\alpha(t)| \leq C \exp(K(t)L\mu([0, t])) \leq C \exp(K(T)L\mu([0, T])).$$

We conclude that $\varphi_\alpha \in L^\infty([0, t_\varphi], \mathbb{R}^n)$.

If $t_\varphi < T$ were true, then statement 1 of Theorem 2.1 would be false, since $\varphi_\alpha(t)$ would lie in the compact set $[0, T] \times \overline{B(0, \|\varphi_\alpha\|_{L^\infty})}$ for every $t \in [0, t_\varphi]$. Since alternative 2 of Theorem 2.1 is trivially false, it follows that $t_\varphi \geq T$.

On the other hand, for $s \leq t$ we have

$$\begin{aligned} |\varphi_\alpha(t) - \varphi_\alpha(s)| &\leq \int_{[s,t)} |f(r, \varphi_\alpha(r))| d\mu \\ &\leq \int_{[s,t)} L\|\varphi_\alpha\|_\infty + \gamma d\mu =: \int_{[s,t)} b(r) d\mu(r). \end{aligned}$$

Note that $b = L\|\varphi_\alpha\|_\infty + \gamma \in L^2([0, T], d\mu)$. The last part of the proposition is an immediate consequence of the previous inequality. $\square$

Definition 4.2. We call the mapping $P : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $P(\alpha) = \varphi_\alpha(T)$, where φ_α is a solution of (IVP), the Poincaré operator.

Solutions to initial value problems associated with MDEs may be discontinuous with respect to time. The following result shows that such solutions depend continuously on the initial condition.

Lemma 4.3. The Poincaré operator is Lipschitz continuous.

Proof. Let φ_α and φ_β be two solutions of (IVP). Since f is Lipschitz, we have

$$\begin{aligned} |\varphi_\alpha(t) - \varphi_\beta(t)| &\leq |\alpha - \beta| + \int_{[0,t)} |f(r, \varphi_\alpha(r)) - f(r, \varphi_\beta(r))| d\mu \\ &\leq |\alpha - \beta| + L \int_{[0,t)} |\varphi_\alpha(r) - \varphi_\beta(r)| d\mu. \end{aligned}$$

Applying Theorem 3.3, we obtain

$$|P(\alpha) - P(\beta)| = |\varphi_\alpha(T) - \varphi_\beta(T)| \leq |\alpha - \beta| e^{K(T)L\mu([0,T])}.$$

This inequality proves the assertion of the lemma. $\square$

Recall the properties of the radius R from the statement of Theorem (1.2).

Theorem 4.4. The closed ball $\overline{B(0, R)}$ is an invariant set for the Poincaré map, i.e. $P(\overline{B(0, R)}) \subset \overline{B(0, R)}$.

Proof. Let $\alpha \in \overline{B(0, R)}$ and let φ_α be the solution of (IVP) defined on a maximal interval. We define the set

$$A = \left\{ t \in [0, T] \mid \varphi_\alpha([0, t]) \subset \overline{B(0, R)} \right\}.$$

We show that the set A is both relatively open and relatively closed in $[0, T]$. Since $[0, T]$ is connected, it follows that $A = [0, T]$.

First, we prove that A is closed. Let (t_n) be a sequence in A that converges to t . If $t_n \geq t$ for some n , then $\varphi_\alpha(s) \in \overline{B(0, R)}$ for every $s \in [0, t]$, and therefore $t \in A$. Conversely, if $t_n < t$ for every n , since φ_α is left-continuous, we have $\lim_{n \rightarrow \infty} \varphi_\alpha(t_n) = \varphi_\alpha(t)$; and, since $\varphi_\alpha(t_n) \in \overline{B(0, R)}$, it follows that $\varphi_\alpha(t) \in \overline{B(0, R)}$, i.e. $t \in A$. Thus, A is closed.

We observe that, by definition, A is an interval. If we suppose that A is not open relative to $[0, T]$, then there exists $t_0 \in A$ such that for every $t > t_0$ it holds that $t \notin A$. Since $t_0 \in A$, we have $|\varphi_\alpha(t_0)| \leq R$. We consider three cases:

Case 1. Suppose that $\varphi_\alpha(t_0) \in B(0, R)$ and $\mu(\{t_0\}) = 0$. Then, by Proposition 4.1, φ_α is continuous at t_0 . Hence, there exists an interval $[t_0, t_0 + \delta)$ such that $\varphi_\alpha([t_0, t_0 + \delta)) \subset B(0, R)$, which contradicts the fact that t_0 is the supremum of A .

Case 2. Suppose that $\varphi_\alpha(t_0) \in \partial B(0, R)$ and $\mu(\{t_0\}) = 0$. According to (H_{43}) , $\beta := -f(t_0, \varphi_\alpha(t_0)) \cdot \varphi_\alpha(t_0)/|\varphi_\alpha(t_0)| > 0$. By Proposition 4.1 and (H_3) , there exists $\delta > 0$ such that if $t \in [t_0, t_0 + \delta)$, then

$$|\varphi_\alpha(s) - \varphi_\alpha(t_0)| < \frac{\beta}{3L} \quad \text{and} \quad |f(s, \varphi_\alpha(t_0)) - f(t_0, \varphi_\alpha(t_0))| \leq \frac{\beta}{3}.$$

Therefore, using the Lipschitz condition on f ,

$$\begin{aligned} [\varphi_\alpha(t) - \varphi_\alpha(t_0)] \cdot \frac{\varphi_\alpha(t_0)}{|\varphi_\alpha(t_0)|} &\leq \int_{[t_0, t)} |f(s, \varphi_\alpha(s)) - f(s, \varphi_\alpha(t_0))| \, d\mu \\ &\quad + \int_{[t_0, t)} |f(s, \varphi_\alpha(t_0)) - f(t_0, \varphi_\alpha(t_0))| \, d\mu \\ &\quad + \int_{[t_0, t)} f(t_0, \varphi_\alpha(t_0)) \cdot \frac{\varphi_\alpha(t_0)}{|\varphi_\alpha(t_0)|} \, d\mu \leq -\frac{\beta}{3} \mu([t_0, t)). \end{aligned}$$

Considering the first-order Taylor expansion of the Euclidean norm around $|\varphi_\alpha(t_0)|$, and taking into account Proposition 4.1, the Cauchy–Schwarz inequality, and the fact that $b \in L^2$, we have

$$\begin{aligned} |\varphi_\alpha(t)| - R &= [\varphi_\alpha(t) - \varphi_\alpha(t_0)] \cdot \frac{\varphi_\alpha(t_0)}{|\varphi_\alpha(t_0)|} + \mathcal{O} \left(\int_{[t_0, t)} b(s) \, d\mu \right)^2 \\ &\leq -\frac{\beta}{3} \mu([t_0, t)) + \mathcal{O} \left(\mu([t_0, t)) \int_{[t_0, t)} b^2(s) \, d\mu \right). \end{aligned}$$

Since $\lim_{t \searrow t_0} \mu([t_0, t)) = \mu(\{t_0\}) = 0$, the right-hand side is negative for t sufficiently close to t_0 . Therefore, there exists a neighborhood V of t_0 such that $|\varphi_\alpha(t)| \leq R$ for all $t \in V$, i.e. A is open. This completes the proof of this case.

Case 3. $\mu(\{t_0\}) \neq 0$. Using the Lebesgue Dominated Convergence Theorem and (H_{42}) , we obtain

$$\lim_{t \searrow t_0} \varphi_\alpha(t) = \varphi_\alpha(t_0) + f(t_0, \varphi_\alpha(t_0))\mu(\{t_0\}) \in B(0, R).$$

Therefore, there exists $\delta > 0$ such that $|\varphi_\alpha(t)| < R$ for all $t \in [t_0, t_0 + \delta)$, contradicting the fact that t_0 is the supremum of A .

Then $A = [0, T]$. Consequently, for any $\alpha \in \overline{B(0, R)}$, it holds that $P(\alpha) = \varphi_\alpha(T) \in \overline{B(0, R)}$. $\square$

We recall the well-known Brouwer Theorem; see [14].

Theorem 4.5 (Brouwer Theorem). *Let $B(0, R)$ be a ball in $\mathbb{R}^n$ and let $P : \overline{B(0, R)} \rightarrow \overline{B(0, R)}$ be a function. If P is continuous, then there exists $x \in \overline{B(0, R)}$ such that $P(x) = x$.*

Now, Theorem 1.2 follows immediately.

Proof of Theorem 1.2. From Lemma 4.3, the operator P is continuous, and by Theorem 4.4, we have $P\left(\overline{B(0, R)}\right) \subset \overline{B(0, R)}$. Applying Theorem 4.5, there exists a solution φ to problem (1.1) such that $\varphi(T) = \varphi(0)$. $\square$

Remark 4.6. Clearly, the condition $f \cdot x < 0$ when $\mu(\{t\}) = 0$ ensures that the flow of the differential equation cannot cross $\partial B(0, R)$ continuously from inside to outside. Condition (H_{43}) prevents an impulse from pushing the solution out of the ball in the same direction.

Remark 4.7. The ball in Theorem 1.2 can be replaced by any set that satisfies Brouwer's Theorem, in particular by any compact and convex set K . In that case, the inequality $f \cdot x < 0$ in Theorem 1.2 must be replaced by $f \cdot \vec{n} < 0$, where $\vec{n}$ is an outward normal vector to K at x . There is another condition that trivially prevents the flow from crossing ∂K continuously; namely, when K is invariant under the flow. These conditions can be combined, which leads to the following result.

Theorem 4.8. *The conclusion of Theorem 1.2 remains true if we replace the ball $B(0, R)$ by a compact and convex set $K \subset \mathbb{R}^n$ and condition (H_{43}) by:*

(H'_{43}) $\partial K = \Gamma_1 \cup \Gamma_2$, where f satisfies $f(t, x) \cdot \vec{n} < 0$, for every (t, x) with $x \in \Gamma_1$ and $\mu(\{t\}) = 0$, and the set Γ_2 is invariant under the flow. That is, if $\mu(\{t\}) = 0$ and $\alpha \in \Gamma_2$, then $\varphi_\alpha(s) \in \Gamma_2$ for every $s \geq t$ sufficiently close to t .

5. AN EXAMPLE: FROUD-ZHUKOVSKY PENDULUM

5.1. Theoretical considerations. In order to illustrate the application of our main theorem, we present a concrete example. We consider the following 2π -periodic boundary value problem:

$$(5.1) \quad \begin{cases} d\theta' = (-\sin \theta - c\theta') d\lambda + g(t, \theta, \theta') d\mu, \\ \theta(0) - \theta(2\pi) = \theta'(0) - \theta'(2\pi) = 0. \end{cases}$$

Recall that λ denotes the Lebesgue measure. This equation corresponds to a damped pendulum subject to impulsive forcing, with viscosity constant equal to c . In the literature, this type of pendulum is sometimes referred to as a Froud-Zhukovsky pendulum (see [28, VI, 4]). We assume that the external force is impulsive and given by a finite sum of Dirac measures:

$$\mu = \sum_{i=1}^n \alpha_i \delta_{t_i},$$

where $D = \{t_1, \dots, t_n\}$ is a finite subset of $[0, 2\pi]$. We assume that $\alpha_i \in \mathbb{R}$, so that μ may be a signed measure. The function g is assumed to be sufficiently regular.

We can rewrite problem (5.1) as

$$dx = f(t, x) d\nu,$$

where ν is the signed measure $\nu = \lambda + \mu$, $x = (x_1, x_2) = (\theta, \theta')$, and

$$f(t, x) = \chi_{[0, T] \setminus D}(t) \begin{pmatrix} x_2 \\ -\sin x_1 - cx_2 \end{pmatrix} + \chi_D(t) \begin{pmatrix} 0 \\ g(t, x) \end{pmatrix}.$$

Here χ_A denotes the indicator function of the set A .

We take the set K as

$$K = \left\{ x \mid -\pi \leq x_1 \leq \pi, \frac{x_2^2}{2} - \cos x_1 \leq 1 \right\}.$$

In physical terms, K is the set of states with energy less than or equal to 1. Moreover, K is closed and convex. In order to satisfy condition (H_{42}) , we assume that $g \equiv 0$ on ∂K . This assumption alone is not sufficient, and we will impose further restrictions on the function g below.

We now show that the hypotheses of Theorem 4.8 are satisfied. We take $\vec{n} = (\sin x_1, x_2)$ and $t \in [0, 2\pi]$ with $\nu(\{t\}) = 0$, that is, $t \notin D$. We first consider the damped case, namely $c > 0$. In this case, the inequality $f \cdot \vec{n} < 0$ holds for $x \in \Gamma_1 := \partial K \setminus \{(\pm\pi, 0)\}$. The set $\Gamma_2 = \{(\pm\pi, 0)\}$ is trivially invariant under the flow, since it consists of equilibrium points. In the undamped case ($c = 0$), the set ∂K is invariant. This follows immediately from an energy conservation argument and from the fact that $c = 0$ and $g = 0$ on ∂K . Therefore, hypothesis (H'_{43}) in Theorem 4.8 is satisfied.

In order for f to satisfy (H_{42}) , we must choose g appropriately. For instance, if g satisfies

$$|g(t, x)| \leq \frac{d(x, \partial K)}{M},$$

where $M = \sup_{t \in D} \nu(\{t\})$, then condition (H_{42}) holds. Under all the above assumptions, problem (5.1) admits at least one solution.

5.2. An algorithm. The shooting method can be used as an algorithm to numerically solve periodic problems associated with MDEs. We implemented this algorithm in the Python programming language¹. It numerically computes Lebesgue–Stieltjes integrals and solves MDEs using Picard’s method of successive approximations. The convergence of the latter procedure was established in [5]. This approach is used to obtain the Poincaré map P .

We then plot contour lines and density plots of the function

$$h(x) = \|x - P(x)\|^2.$$

This type of visualization provides insight into the regions where it is convenient to search for fixed points of P , and may also reveal all such fixed points. Finally, the fixed point problem $P(x) = x$ is solved using optimization routines from the module `scipy.optimize`.

Successive approximations are perhaps not the most efficient algorithm for solving MDEs; our intention is to study numerical methods for MDEs in greater depth in future work.

Returning to the particular problem (5.1), we consider two experiments satisfying all the above hypotheses. Therefore, Theorem 1.2 guarantees the existence of periodic solutions. We are interested in finding these solutions numerically and interpreting them in physical terms.

¹<https://github.com/fdmazzone/shootingMDE>

Experiment 1: damped pendulum. We take $c = 2.0$, $D = \{1, \dots, 5\}$, $\mu = \sum_{j=1}^5 (-1)^j \delta_j$ and $g(t, x) = \max\{1.9 - \|x\|, 0\}$. Then $M = 1$, and it is not hard to see that g verifies the inequality $|g| \leq d(x, \partial K)$. In Figure 1 the density plot of $h(x) = \|x - P(x)\|^2$ is drawn.

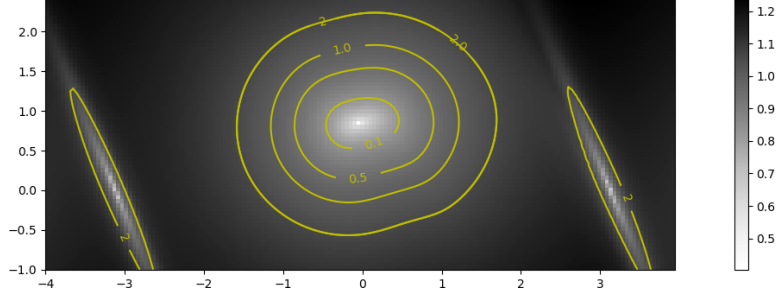


FIGURE 1. Level sets of $h(x) = \|x - P(x)\|^2$, $c = 2.0$, $\mu = \sum_{j=1}^5 (-1)^j \delta_j$.

The density map presumably shows that there are only three periodic solutions. Two of them correspond to the unstable equilibria $x = (\pm\pi, 0)$ of the simple pendulum, which continue to be equilibria of our equation. Note that the stable equilibrium $(0, 0)$ of the simple pendulum is no longer an equilibrium of our equation, since the impulsive forces alter its steady state. This is not true for unstable equilibria, since $g = 0$ in ∂K . A third solution can be seen in the center of the image. Its position $(-0.04, 0.85)$ was determined more accurately using the optimization algorithm. In Figure 2, we plot the corresponding solution. As can be observed, in this case the mass spends most of its motion at negative angles. The impulsive forces practically do not allow the mass to cross the equilibrium position $\theta = 0$.

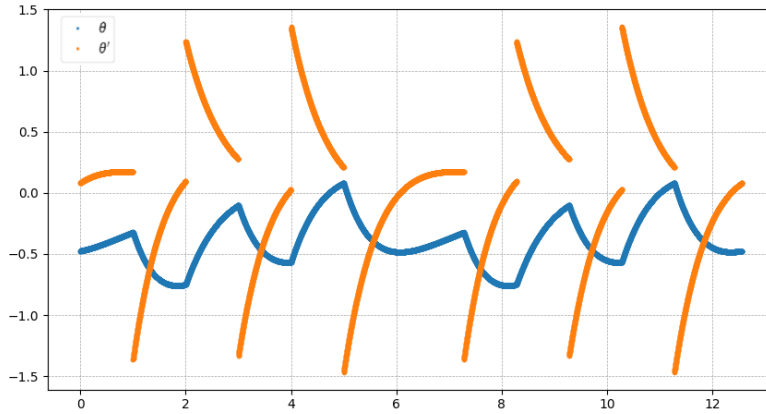


FIGURE 2. Graph of the solution (θ, θ') of (1.1) on the interval $[0, 4\pi]$, with $c = 2.0$ and $\mu = \sum_{j=1}^5 (-1)^j \delta_j$.

Experiment 2: undamped pendulum. In this example, we take $c = 0.0$, $D = \{1, \dots, 5\}$ (again), and $\mu = 0.5 \sum_{j=1}^5 \delta_j$. The function g is the same as in the previous example. In this case, we are interested in understanding what a periodic solution looks like when the impulsive forces always act in the same direction.

In Figure 3, the density plot of $h(x) = \|x - P(x)\|^2$ is shown. The plot clearly displays the compact region K and the solutions of the equation $h(x) = 0$. The unstable solutions $(\pi, 0)$ and $(-\pi, 0)$ lie on the boundary of K , where the pendulum behaves like a simple pendulum because the external force g vanishes on ∂K .

As in the previous example, the stable equilibrium $(0, 0)$ of the simple pendulum is no longer an equilibrium point of our equation. We observe a solution of $h(x) = 0$ inside K , at the point $P = (1.338, 0.578)$, which corresponds to the periodic solution we were looking for and was obtained through an optimization algorithm. Its graph is shown in Figure 4.

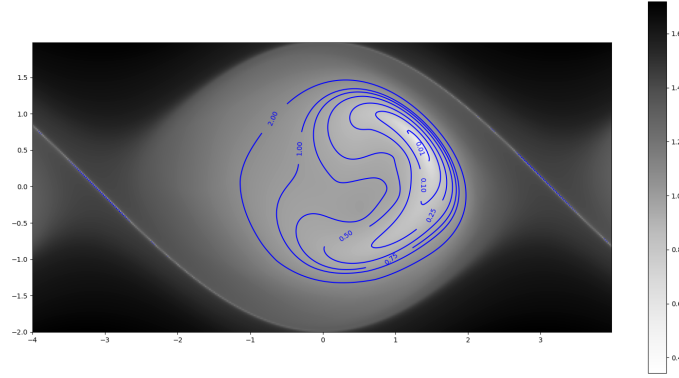


FIGURE 3. Level sets of $h(x) = \|x - P(x)\|^2$, with $c = 0.0$ and $\mu = 0.5 \sum_{j=1}^5 \delta_j$.

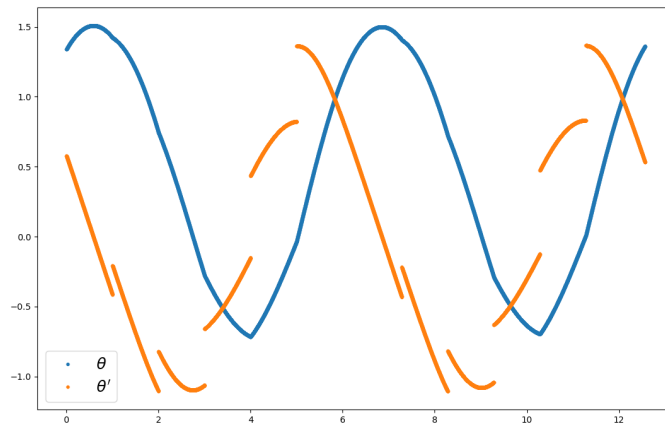


FIGURE 4. Graph of the solution (θ, θ') of (1.1) on the interval $[0, 4\pi]$, with $c = 0.0$ and $\mu = 0.5 \sum_{j=1}^5 \delta_j$.

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